

JAVA STRINGS PRACTICE QUESTIONS

1. Take a full name as input and print:

- Original name
- Length
- Uppercase version

Topics: `length()`, `toUpperCase()`, methods

2. Print the first, middle, and last character of a string.

Topics: `charAt()`, `length()`, if-else

3. Count the number of vowels, consonants, digits, and spaces in a string.

Topics: `charAt()`, if-else, loops

4. Reverse a string without using any built-in reverse method.

Topics: `charAt()`, loops, String concatenation

5. Check whether a string is a palindrome.

Topics: `charAt()`, `equals()`, loops, if-else

6. Count how many times a specific character appears in a string.

Topics: `charAt()`, loops, if-else

7. Remove all spaces from a string.

Topics: `replace()`, loops (optional)

8. Replace all vowels in a string with '*'.

Topics: `charAt()`, if-else, `StringBuilder`

9. Extract the username and domain from an email address.

Example: nishanth@gmail.com

Topics: indexOf(), substring()

10. Check whether a string starts with a given prefix and ends with a given suffix.

Topics: startsWith(), endsWith(), if-else

11. Count the number of words in a sentence.

Topics: charAt(), loops, if-else

12. Find the longest word in a sentence.

Topics: substring(), charAt(), loops, if-else

13. Convert the first letter of every word to uppercase.

Example:

"java full stack" → "Java Full Stack"

Topics: charAt(), substring(), toUpperCase(), loops

14. Compare two strings using both == and equals(), then explain the difference.

Topics: ==, equals(), String Pool

15. Create two strings using new String() and predict the output of == and equals().

Topics: Objects, ==, equals()

16. Check whether two strings are equal ignoring case differences.

Topics: equalsIgnoreCase(), if-else

17. Determine whether a string contains only digits.

Topics: `charAt()`, if-else, loops

18. Validate a password:

- Minimum 8 characters
- At least one uppercase letter
- At least one digit

Topics: `length()`, `charAt()`, if-else, loops

19. Reverse every word in a sentence while keeping word order the same.

Example:

"Java Full Stack" → "avaJ lluF kcatS"

Topics: `substring()`, `charAt()`, loops, `StringBuilder`

20. Count the frequency of each vowel in a string.

Topics: `charAt()`, if-else, loops

21. Create a `StringBuilder` and perform:

- `append()`
- `insert()`
- `delete()`
- `replace()`
- `reverse()`

Topics: `StringBuilder` methods

22. Generate an employee ID using `StringBuilder`.

Example:

EMP2026001

Topics: `StringBuilder`, `append()`

23. Compress a string.

Example:

aaabbccdd → a3b2c2d2

Topics: charAt(), StringBuilder, loops, if-else

24. Check whether two strings are anagrams.

Example:

listen and silent

Topics: Arrays, sorting, equals()

(Keep this for later after Arrays)

25. Build a mini user-registration system:

- Trim spaces
- Convert username to lowercase
- Validate email contains '@'
- Validate password length

Topics: trim(), toLowerCase(), contains(), length(), if-else, methods